Measure, Monitor, Act, Display

Introduction

The main purpose of this system is to monitor sensor readings and take an action when a certain condition is reached. The system is based on an Arduino UNO, which receives measurements from a voltage input and an MQ-2 gas sensor. When the measured voltage reaches 8 V, when the gas level reaches the required limit, or when a manual command is given through the MATLAB GUI, the system activates a buzzer and a DC motor. The Arduino therefore works as the main controller that reads the sensors, processes the readings, and controls the output devices. This type of system combines sensing, signal conversion, processing, and communication.

Analog-to-Digital Converter (ADC)

An Analog-to-Digital Converter (ADC) is a circuit that converts an analog signal, such as a voltage that can have many possible values, into a digital value that a microcontroller can process. Sensors normally produce analog signals, while a microcontroller works with digital numbers. The ADC acts as the connection between the two.

The conversion process mainly depends on sampling and quantization. Sampling means taking measurements of the analog signal at specific points in time, while quantization maps the measured voltage to one of a limited number of digital levels. The number of available levels depends on the number of ADC bits. For example, an N-bit ADC has 2^N possible digital levels. ADC performance can be described using parameters such as sampling rate, number of bits, quantization levels, and dynamic range.

For this project, the ADC is important because the Arduino needs to convert the voltage coming from the sensing circuit into a numerical value. The Arduino can then compare this value with the required threshold and decide whether the alarm and motor should be activated.

Arduino UNO ADC

The Arduino UNO uses the ATmega328 microcontroller, which contains a built-in ADC. The ADC allows the analog input pins to be used for measuring voltage and converting the measurement into a digital value. The ADC can then be accessed through the Arduino development environment, making it possible to read sensor signals in the Arduino program. The ADC operation of the ATmega328 and its implementation on the Arduino UNO are discussed in the selected reference.

In this project, the Arduino ADC is used to measure the input voltage. Since the system is required to measure a voltage ranging from 0 to 12 V, the voltage cannot simply be connected directly to an Arduino analog input. A voltage-divider circuit is therefore required to reduce the voltage to a safe range before it reaches the ADC.

MQ-2 Gas Sensor

The MQ-2 is a gas sensor commonly used for detecting smoke and several combustible gases. Its output changes depending

on the surrounding gas concentration, allowing a microcontroller to use its signal to determine whether the gas or smoke level has reached a specified condition.

The MQ-2 can be connected to a microcontroller as part of an automatic detection system. One of the selected studies demonstrates an indoor smoke detection system using an MQ-2 sensor and an Arduino UNO. In that system, when smoke is detected, the Arduino activates a buzzer and an exhaust fan. This shows how an MQ-2 sensor can be used as the sensing part of an automatic control system.

For the EURUS system, the MQ-2 provides another input to the Arduino. The Arduino reads the sensor signal and compares it with the selected gas-level condition. If the required level is reached, the system activates the alarm and DC motor.

Potential Divider

A potential divider, also called a voltage divider, is a simple circuit made from two resistors. It is used to obtain a lower voltage from a higher input voltage.

The output voltage is given by:

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

where V_{in} is the original input voltage and V_{out} is the reduced voltage measured by the Arduino.

The potential divider is necessary in this project because the system has to measure an input voltage from 0 to 12 V, while the Arduino ADC input must receive a voltage within its allowed input range. The divider reduces the 0–12 V signal to a suitable level before it reaches the Arduino ADC. This allows the Arduino to measure the input without applying the full voltage directly to the microcontroller.

Signal-conditioning circuits are an important part of connecting real-world sensors to ADCs because the signal may need to be changed into a suitable voltage range before conversion.

Overall System Operation

The main idea of the system can be summarized as:

Sensor/Input → Signal Conditioning → Arduino ADC → Arduino Processing → Output/Communication

The voltage input first passes through the potential divider, which reduces its level before it is measured by the Arduino ADC. The MQ-2 provides a separate sensor input related to gas or smoke detection. The Arduino converts the analog readings into digital values and compares them with the programmed conditions.

If the voltage reaches **8 V**, if the gas level reaches the required threshold, or if the user activates the system manually through the MATLAB App, the Arduino activates the buzzer and DC motor. The MATLAB App is connected to the Arduino through a wired connection and provides a way to monitor the system and manually trigger the output when needed.

This combination of sensing, ADC conversion, processing, actuation, and communication forms the basic architecture of the proposed monitoring and control system.